

Project 66: Real-Time Audio Transcriber

Goal: Capture audio from a microphone and **transcribe it live** using a speech-to-text engine like:

-  **OpenAI Whisper** (local/offline, accurate)
 -  **Google SpeechRecognition** (fast, easy)
 -  **Vosk / DeepSpeech** (offline, lightweight)
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Tools You Can Use

Engine

Pros

Cons

Whisper

Very accurate, offline

Slower, needs GPU for real-time

SpeechRecognition + Google

Fast, easy

Requires Internet

Vosk

Fast, offline

Medium accuracy

Step-by-Step: Using speech_recognition (Simple Setup)

Step 1: Install Required Packages

```
pip install SpeechRecognition pyaudio
```

If pyaudio fails, use:

```
pip install pipwin  
pipwin install pyaudio
```

Step 2: Create Real-Time Transcriber

```
import speech_recognition as sr  
  
recognizer = sr.Recognizer()  
  
def live_transcribe():  
    with sr.Microphone() as source:  
        print("🎤 Listening... (say 'stop' to end)")  
        while True:  
            try:  
                audio = recognizer.listen(source)  
                text = recognizer.recognize_google(audio)  
                print("📝 ", text)  
                if "stop" in text.lower():  
                    print("🛑 Stopping transcription.")  
                    break  
            except sr.UnknownValueError:  
                print("🗣️ Couldn't understand. Try again.")  
            except sr.RequestError as e:  
                print("⚠️ API error:", e)
```

Run it:

```
live_transcribe()
```

Output

Live stream of printed transcriptions, ending when you say “stop”.

Whisper Alternative (Advanced + Accurate)

Install:

```
pip install openai-whisper sounddevice numpy
```

Use Whisper + SoundDevice to stream audio and transcribe chunks every few seconds.

Project Summary

- **Input:** Real-time audio via microphone
- **Output:** Live transcription printed or displayed
- **Tool:** `speech_recognition` or `whisper`
- **Use Case:** Live captions, meetings, classrooms, accessibility